6COSC023W

Computer Science Final Project

Project Proposal and Requirements Specification (PPRS) Handout

HeartSmart

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Declaration

This document has been prepared based on my own work. Where other published and unpublished source materials have been used, these have been acknowledged in references.

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1. Definition of the Problem

- **Problem Description**: Heart Disease, specifically, Ischemic heart disease, is the leading cause of death in the world. The main underlying condition for this being Coronary Artery Disease. Coronary Artery Disease is influenced by modifiable lifestyle factors such as exercise, diet, smoking and more. Despite the preventable nature of Coronary Artery Disease individuals remain unaware of their risk for the disease and how they live their lives affects it. But some individuals do act using Heart Health related applications. However, you'll find that current applications are either too focused on surface level heart-related metrics or are too intimidating when it comes to their delivery or are just generally less concerned in educating the user and ensuring they do not forget what they learned in order to take preventive action.

  This project will aim to bridge this gap between education and aid by still helping assess individuals' risks whilst making sure to educate individuals on Coronary Artery Disease so users can take action against such a deadly disease that kills so many people every year.
- **Significance**: This project combines Machine Learning/Data Processing, Health/Medical Information and Application Development to help individuals fight against and prevent a disease that claims the most lives year on year. By applying predictive modelling from Machine Learning/Data Science it will hold the utmost significance in Machine Learning led healthcare/wellbeing.

2. Aims, Objectives, and Scope

- **Aim**: The primary goal of this project is to develop an application that can access an individual's risk towards Coronary Artery Disease and educate the individual in preventing such a disease through lifestyle choices such as diet, exercise, smoking and more. The project also aims to address the gap in other applications by properly educating individuals through subtle gamification, allowing them to not only learn but apply their knowledge in quizzes and mini tests to ensure they are actively recalling said knowledge which ensures it sticks with them forever.

- **Objectives**:
    - Develop a user-friendly interface that allows users to input certain measurements/data to predict and assess their risk against Coronary Artery Disease, this can include but is not limited to Cholesterol, Age, Sex, Weight, Smoking Status and more.
    - Educational section which will allow users to learn facts and apply it through quizzes/mini tests to ensure active recall.
    - Source and process an anonymous heart disease dataset.
    - Create/train a Machine Learning/Data Science predictive model (based on available datasets/databases) that predicts the risk an individual is at for Coronary Artery Disease. This model will predict this risk based on the information the user provides. The models performance will be judged through machine learning performance metrics.
    - Integrate features such as tracking diet, exercise and possible medication use.
    - Ensure the application is tailored to all types of people and helps them fight against Coronary Artery Disease.
    - Provide reminders to update heart/medical related /data to predict risk of Coronary Artery Disease every 3 months.
- **Scope**:
    - **Inclusions**:
    - A Coronary Artery Disease Risk Prediction/Assessment. It will check an individual's risk of Coronary Artery Disease based on several factors such as age, sex, weight, etc.
    - CAD prevention via diet, exercise or medicine tracking. Allows user to be on top of things through their lifestyle.
    - Coronary Artery Disease information/facts with quizzes through educational section.
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    - **Exclusions**:
    - Coronary Artery Diagnosis/Treatment, this application should not be used as a substitute for a real professional medical opinion/diagnosis.
    - Ability to integrate wearable devices such as watches and or medical equipment to monitor and take certain measurements/data in real time.

3. Background Review

- **Key Findings (Literature / Systems)**:
    - Research shows that integration of emerging technologies is driving a big revolution in the health and wellbeing space (Thatcharodi et al, 2024). With this being the case, I wanted to find out how much of a role Machine Learning and Data Science played a role in prevention and diagnoses in the health and wellbeing sector. In the case of prostate cancer, one of the most common types of cancer in men, machine learning is making crucial diagnostic and therapy decisions. Machine Learning can make risk predictions, reducing medical errors and improving productivity (Chen, et al 2024). Not only can Machine Learning algorithms help in decision making, but researchers also believe that can save huge expenses and time needed for medicinal studies (Ahsan et al, 2023). All in all, Ahsan et al found that for Predictive Analytics to analyse patient data and predict patient outcomes, such as probability of contracting a specific disease, Machine Learning can assist in making more informed choices regarding care and treatment for patients (Ahsan et al, 2023). I wanted to focus on the more preventative side of things when it came to health. To be more specific, I wanted to target the disease that claims the most lives year on year. The World Health Organization stated that Ischemic Heart Disease was the leading cause of death worldwide. The biggest underlying factor of this disease being Coronary Artery Disease (CAD). When it came to CAD, I wanted to focus on the risk factors that an individual can control in order to take a preventative approach against Coronary Artery Disease. According to Dai et al, amongst all potentially modifiable risk factors, dietary risks followed by high systolic blood pressure were the leading factors for IHD, 69.2% and 54.4% respectively for causing deaths (Dai et al, 2022). With this being said I wanted to focus on a project for an application that can both address risk assessment through Machine Learning Predictive Modelling as well as educating individuals on the lifestyle changes and choices they make that can either prevent or put them at risk of Coronary Artery Disease (CAD). With my goal being set, I wanted to look further into how effective health and wellbeing applications could be in preventing disease and teaching individuals valuable information they will apply in their day to day life. Deni-Garcia et al have found that if implemented

correctly, mobile health application with Artificial Intelligence can be considerably useful for both individuals and health care professionals in prevention of diseases (Deniz-Garcia et al, 2023). They raised concerns of data privacy, user engagement and behaviour change. I echo this point 100% and believe these applications will have incredibly benefits towards preventative measures for deadly diseases, especially if we address the primary issues. Finally, after all my research as well as having looked at the technology on the market; the gap I plan to fill is an application that address issues of user engagement and behaviour change whilst still attempting to stick to the norm of applications that can help calculate and keep track of an individual's risks of Coronary Artery Disease.

- **Comparison with Similar Software Applications/Products**
  - **CardioSmart360**:
  - Strengths:

    -Incredibly informative on issues related to the Heart (Heart Diseases, Heart Attacks, Heart Anatomy and more). This is something I am to replicate to an extent.

    -Physical Diagrams, Models and Videos on Heart related issues/anatomy. I will likely not do this as I am not creating an anatomical modelling tool.

    Weaknesses:

    -Is largely only informative. Doesn't allow the user to test their knowledge through quizzes which will allow the user to actively recall information and make sure it sticks in their memory. This is something I wish to tackle in my application.

    -Not really related to Coronary Artery Disease prevention through risk assessment, diet, exercise and more. Completely different to what I plan to do.

  - **HeartRate**:
  - Strength:

    -Reminders to take measurements such as Heart Rate, Blood pressure, Blood Oxygen and more. I think reminders are a great idea and it is an idea I will look to include in my own application.

-Pedometer to track movement data. I will not be looking into any real time tracking so this feature is not something I will be looking to emulate.

Weaknesses:

-Not relevant or related to Coronary Heart Disease, it's just a general heart related data tracker. This is a weakness I will tackle as my application is more tailored towards CAD.

-Lacks the educational/information aspect of Heart health. As mentioned before, this is something I will be tackling, aiming to educate individuals.

- **Heartify**:
- **Strengths**:

  -Full of information related to Heart Health such as diet, exercise, risk factors and more. Very educational and is something I will be looking to replicate in my own application in order to educate individuals.

  -Has a Heart Disease Risk Assessment to check our 10-year risk of cardiovascular disease. Allows you to also export Health Report. Another feature-set I will be looking to emulate in my own application. One of my main objectives.

  -Takes measurements and has daily tasks to complete. Another feature I will be looking to implement in my application, it's a great idea and goes hand in hand with my other requirements.

  Weaknesses:

  -Doesn't offer users to test their knowledge through quizzes which would allow them to actively recall information so it can stick longer. This is the main gap I have found in applications and something I will be tackling. Active recall is important to make sure education/information sticks.

  -Certain features stuck behind a paywall. All my features will be free.

- **Differentiation**: My application will mainly aim to bring everything together and take some of the best parts about other applications and combine it into one. Additionally, I have found that applications are lacking in the active recall portion of knowledge.

Active recall is important to remembering information in the long term, it is important that users/individuals are tested to ensure that they do not forget critical information that can help in fighting against and prevent Coronary Artery Disease.

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	Current State (FROM)	Gaps	Future State
Quizzes/Information recall	Current applications without quizzing/testing.	Current applications are lacking quizzing for the knowledge they are teaching.	Applications with quizzing/testing.

4. Tools (Hardware/Software)

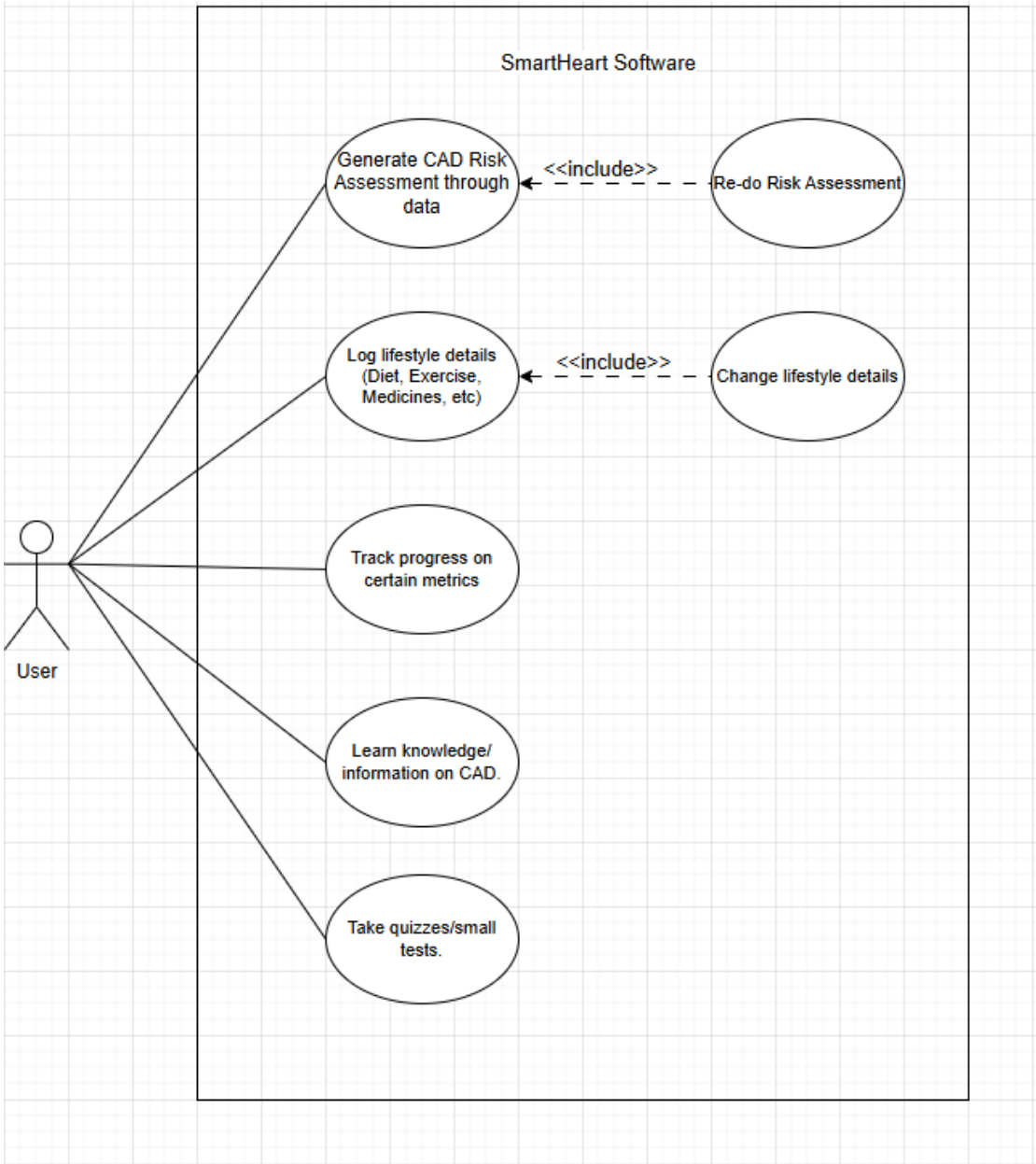
- **Hardware**:
  - Laptop/Desktop
- **Software**:
  - **Python**: Front and backend.
  - **Draw.**io/Diagrams.net: Creating use case and other diagrams.
  - **Jupyter Notebook**: For creating and training machine learning model..
  - **SQLite**: Creating Database for user related information.
  - **VSCode: IDE and workspace for programming and coding.**
  - **GitHub: Version control and code management.**
  - **Streamlit: Front-end**

5. Use Cases and Diagrams

- **Use Cases**:
  - Use Case 1: Allows user to get a Coronary Artery Disease Risk Assessment by inputting certain details.
  - Use Case 2: Allows user to log details such as diet, exercise and medicine usage.
  - Use Case 3: Allows user to track progress on certain metrics.
  - Use Case 4: Allows user to learn new knowledge on Coronary Heart Disease.

- Use Case 5: Allows user to test their knowledge through quizzing/small tests.

- **Diagrams**:



6. Requirements Elicitation

- **Approach**:
  - **[Method 1]**: Academic Research
  - **[Method 2]**: Background Research

- **Techniques**:
- **Use of Academic Articles, Papers, Books and more for requirements gathering.**
- Using similar applications to see what requirements may look like for this application.
- **Researching into Coronary Heart Disease and learning how humans maintain and understand concepts.**
- **Requirement Specification:**

Functional Requirements:

-Input data for risk assessment:

Users can enter their age, sex, weight, height, smoking status, blood pressure, diabetes status and cholesterol. Some of this will be optional as some users may not have access to certain information.

-Coronary Artery Disease Risk Prediction:

Application calculates users' risk for coronary artery disease at a high confidence and explains the major contributing factors towards CAD.

-Logging Lifestyle Information:

Users can create, update or delete logs for diet, exercise, medicine usage and body measurements such as blood pressure and cholesterol to track progress.

-Track Progression:

Users can view charts and graphs that show trends and progress for their blood pressure, cholesterol and other trackable information.

-Update Risk Assessment:

Users can get an updated risk assessment at a set time interval and are encouraged to do so.

-Educational Section:

Users are given facts on heart diseases, diet, exercises and other risk factors and overall useful information of Coronary Artery Disease. Users can also quiz themselves on what they learnt, and the quizzes scores are stored. After quiz attempt, answers are explained and given. Allows for gamification.

-Data export/deletion:

Allows users to export their data for further use elsewhere as well as delete their data.

-Medical Disclaimer:

Warning user that the application isn't a diagnostic tool and should not be a replacement for a real medical opinion and diagnosis.

Non-functional Requirements:

-Complete Risk Assessment with least amount of information provided whilst still being possible.

-Risk Assessment shouldn't be complex and take too many steps. Ideally less than 10 steps.

-Risk Assessment Prediction should be quick, taking less than a second to predict risk.

-Application should load quickly and shouldn't take more than a couple seconds.

-Logs are taken quickly, shouldn't take more than 1 second to log information.

-Application should rarely fail and in that event of a fail the application should have fail safes and failure detection to remedy failures.

-Application should always be up and running and available.

-Predictive Model should perform well against performance metrics to ensure the model is good at assessing risk.

-User data should be encrypted and stored correctly to ensure no data is leaked. Should be complaint with the relevant GDPR and other policies.

7. Time Schedule

- **Gantt Chart Overview**:

8. References / Bibliography

- Include a list of cited/used in your text items (books, papers, websites, etc.). Use the Harvard style for the purpose, or any other preferred standard referencing style.

- **Example Citations**:

  - Thacharodi, A., Singh, P., Meenatchi, R., Ahmed, Z.H.T., Kumar, R.R.S., V, N. et al. (2024) 'Revolutionizing healthcare and medicine: the impact of modern technologies for a healthier future—A comprehensive review', Health Care Science, 3, pp. 329–349. doi:10.1002/hcs2.115
  - Ahsan, M., Khan, A., Khan, K.R., Sinha, B.B. and Sharma, A. (2024) 'Advancements in medical diagnosis and treatment through machine learning: A review', Expert Systems, 41(3), e13499. doi:10.1111/exsy.13499
  - Chen, X., Liu, X., Wu, Y., Wang, Z. and Wang, S.H. (2024) 'Research related to the diagnosis of prostate cancer based on machine learning medical images: A review', International Journal of Medical Informatics, 181, 105279. doi:10.1016/j.ijmedinf.2023.105279
  - Dai, H., Abu Much, A., Maor, E., Asher, E., Younis, A., Xu, Y., Lu, Y., Liu, X., Shu, J. and Bragazzi, N.L. (2022) 'Global, regional, and national burden of ischaemic heart disease and its attributable risk factors, 1990–2017: results from the Global Burden of Disease Study 2017', European Heart Journal – Quality of Care and Clinical Outcomes, 8(1), pp. 50–60. doi:10.1093/ehjqcco/qcaa076
  - Deniz-Garcia, A., Fabelo, H., Rodriguez-Almeida, A.J., Zamora-Zamorano, G., Castro-Fernandez, M., Alberiche Ruano, M.D.P., Solvoll, T., Granja, C., Schopf, T.R., Callico, G.M., Soguero-Ruiz, C. and Wägner, A.M.; WARIFA Consortium (2023) 'Quality,

usability, and effectiveness of mHealth apps and the role of artificial intelligence: Current scenario and challenges', Journal of Medical Internet Research, 25, e44030. doi:10.2196/44030

9. Ethics Form

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